

Affordable, Portable, Real-Time Chlorophyll Sensor

Physics Report | Group 04

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Executive Summary

This report details the development of a portable, high-accuracy chlorophyll sensor utilizing **Laser-Induced Fluorescence (LIF)** and **tinyML** algorithms. Designed for agricultural, herbal, and aquatic monitoring, the system provides real-time, calibrated measurements of both Chlorophyll-a (Chl-a) and Chlorophyll-b (Chl-b) concentrations.

1 Introduction

1.1 Importance of Chlorophyll

Chlorophyll is a primary indicator of plant health, product quality in the herbal industry, and algal biomass in aquatic ecosystems. Monitoring its concentration is critical for water quality assessment and precision agriculture.

1.2 Detection Methodology

While traditional methods (HPLC, Mass Spectrometry) offer high precision, they are expensive and laboratory-bound. Existing affordable meters often provide only relative content and lack absolute calibration.

- **Research Gap:** Creating an affordable sensor providing calibrated, simultaneous measurements of Chl-a and Chl-b.
- **Objective:** Prototype a real-time sensor using LIF and on-device edge computing analysis.

2 Theoretical Framework

2.1 Physics of Laser-Induced Fluorescence (LIF)

The sensor employs a flow-through system where samples are excited by targeted wavelengths:
* **Laser 405 nm:** Optimized for Chl-a excitation. * **Laser 450 nm:** Optimized for Chl-b excitation.

2.2 Spectral Red Shift & Concentration Features

A critical physical observation in this study is the **Red Shift** of the fluorescence peak as concentration increases. This phenomenon occurs due to:
* **Inner Filter Effect (IFE):** Re-absorption of the shorter-wavelength side of the emission spectrum by the chlorophyll matrix.
* **Self-Absorption:** The overlap between chlorophyll absorption and emission bands causes selective re-absorption at shorter wavelengths.
* **Concentration Quenching:** At high concentrations, molecular aggregation further shifts the emission to longer wavelengths.

These effects produce three measurable spectral features that vary systematically with concentration, as confirmed by our simulation study:

Unlike static peak-height models, our PLS and SVR models analyze the **entire spectral curve**, allowing them to correlate this non-linear shift with absolute concentration. Importantly, our

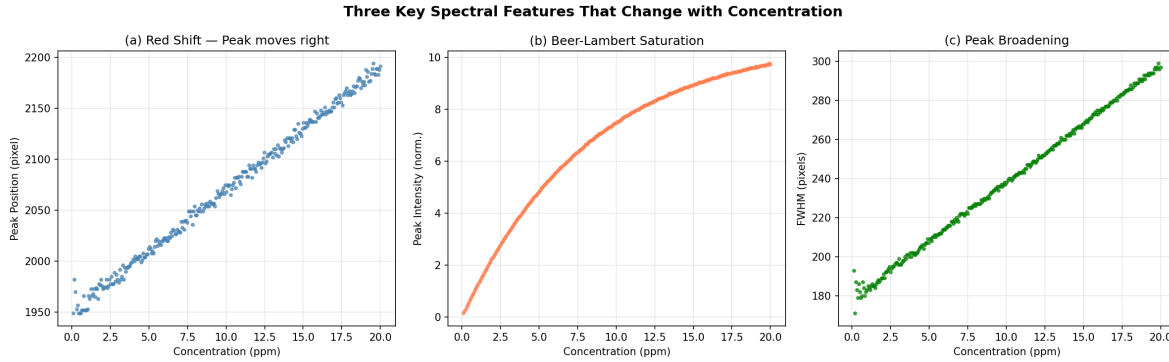


Figure 1: Three key spectral features that change with chlorophyll concentration: (a) peak position shifts ~ 240 pixels across the 0–20 ppm range, (b) intensity saturates following Beer-Lambert law, and (c) FWHM broadens linearly.

simulation study confirmed that **retaining the red shift** (i.e., not aligning peaks) provides superior model performance compared to peak-aligned spectra.

3 Experimental Setup & Hardware

3.1 System Architecture

- **Control Unit:** STM32H743VIT6 (High-performance ARM Cortex-M7 running at 480MHz) for high-speed acquisition and on-device tinyML inference.
- **Sensor:** Toshiba TCD1304 linear CCD with 3648 active pixels, providing high spatial resolution for fluorescence spectroscopy.
- **Optical Path:**
 - **Excitation:** Dual-laser system (405 nm and 450 nm) targeting the Soret bands of Chl-a and Chl-b, alternately activated via servo-controlled beam positioning.
 - **Dispersion:** 600 lines/mm holographic diffraction grating to resolve the fluorescence emission spectrum.
 - **Collimation:** Cylindrical lens and convex mirror for beam direction; objective lens alignment is performed manually prior to acquisition.
- **PCB Design:** A custom PCB was developed using **KiCad** and manufactured via **JLCPCB** to ensure signal integrity and portability.

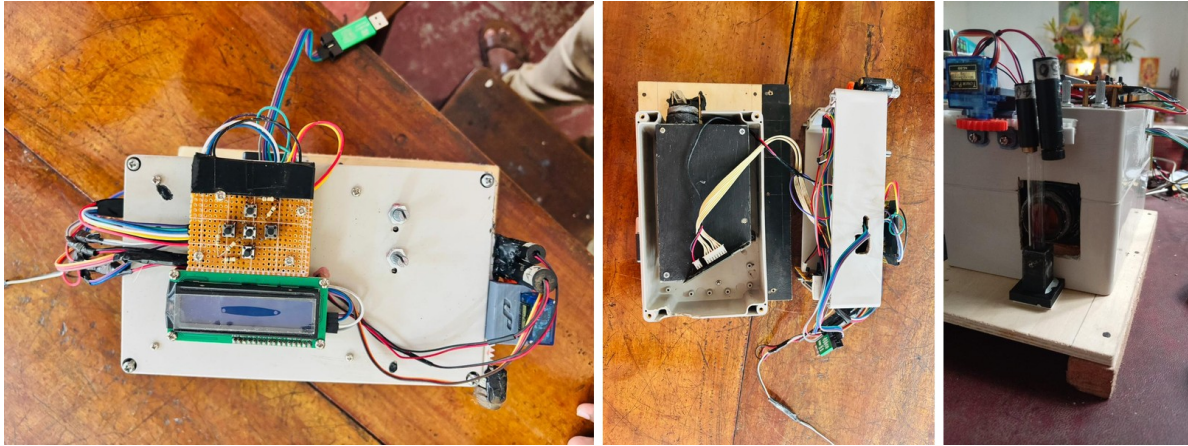


Figure 2: Instrument overview — (left) top-down view of the assembled optical bench, (centre) interior showing the diffraction grating and CCD mount, (right) front panel with cuvette holder and I/O connections.

3.2 System Connectivity

The following diagram illustrates the physical and logical connections between the core components:

3.3 Information Pipeline

The data flow from raw photons to on-device concentration prediction is summarized below:

3.4 Circuitry and PCB Design

The system's electronic backbone is a custom-designed PCB optimized for low-noise CCD readout and high-speed signal processing.

Note: The full vector PCB schematic is available in the electronic HTML version of this report. Refer to Figure 5 for the physical board implementation.

3.4.1 Key Hardware Features

- **MCU High-Speed Interfacing:** The STM32H7 communicates with the TCD1304 CCD using high-speed timers to generate the Master Clock (fM) and Shift Gate (SH) pulses, ensuring synchronized pixel acquisition.

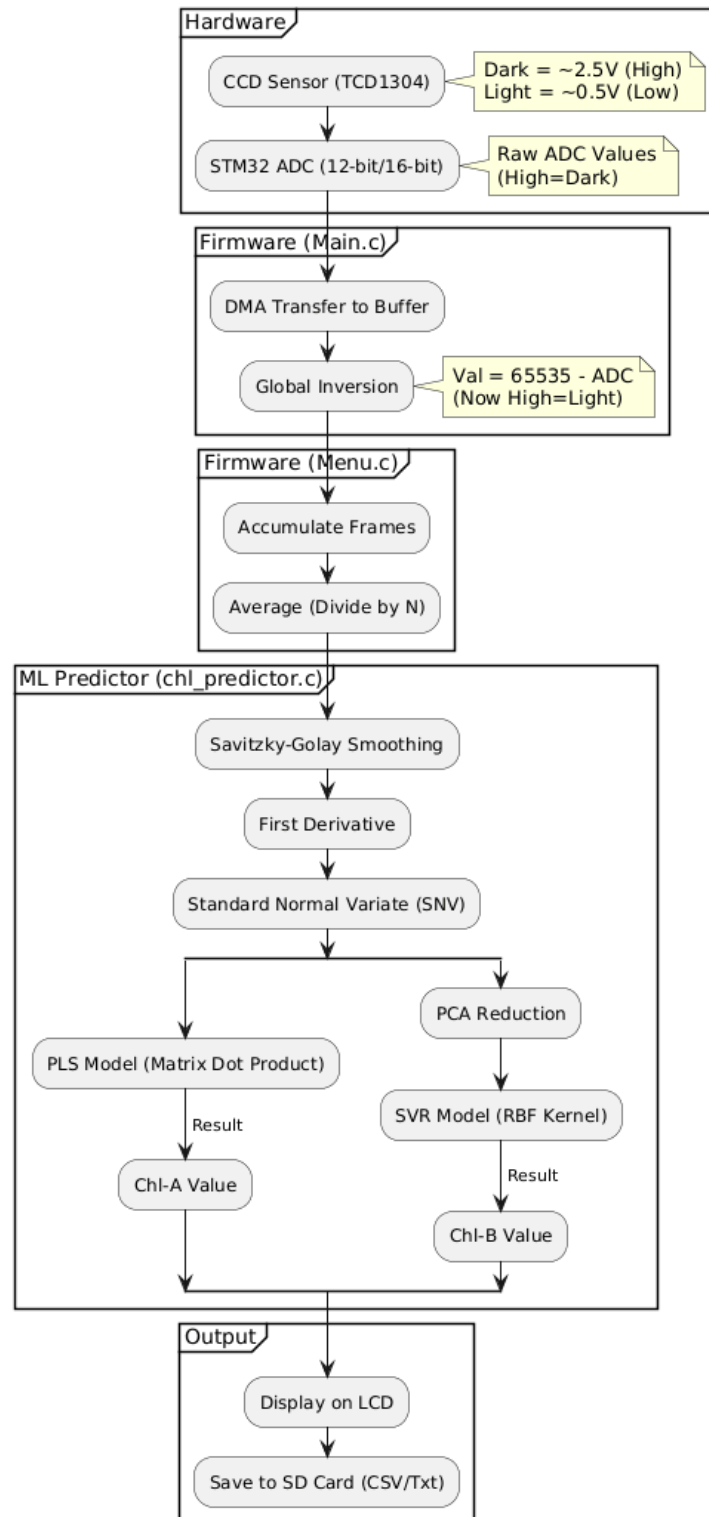


Figure 3: System connectivity diagram showing physical and logical connections between STM32H743VIT6, TCD1304 CCD, lasers, servo, LCD, SD card, and USB

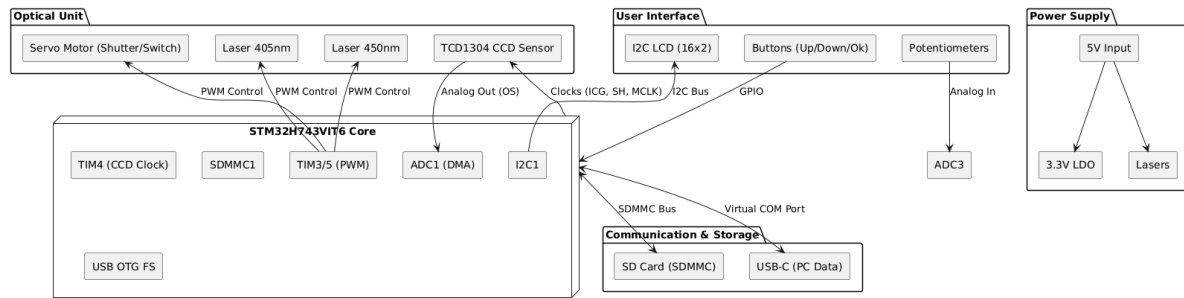
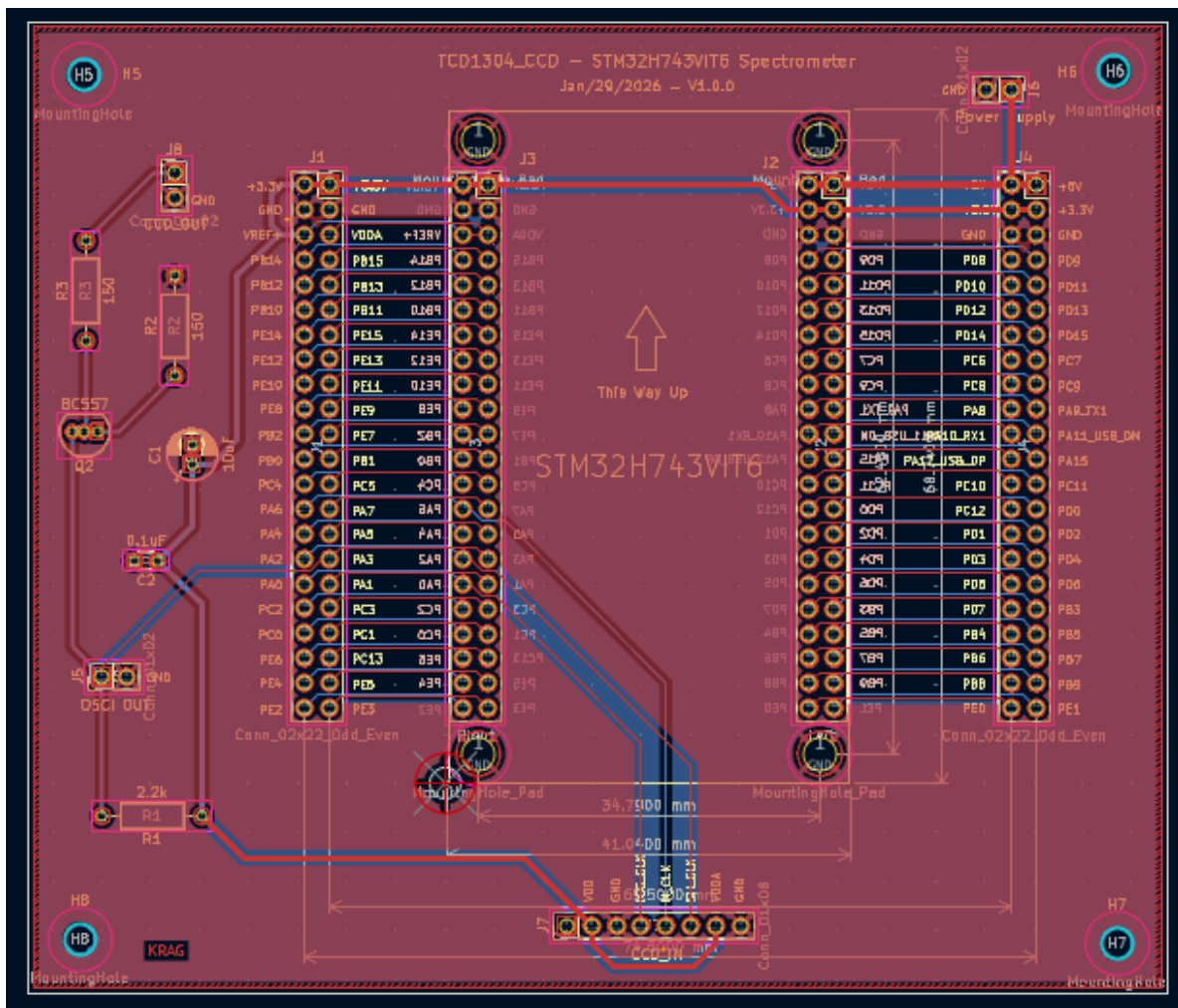


Figure 4: Software architecture class diagram showing the modular firmware design with Core Application, Signal Processing, and Storage packages



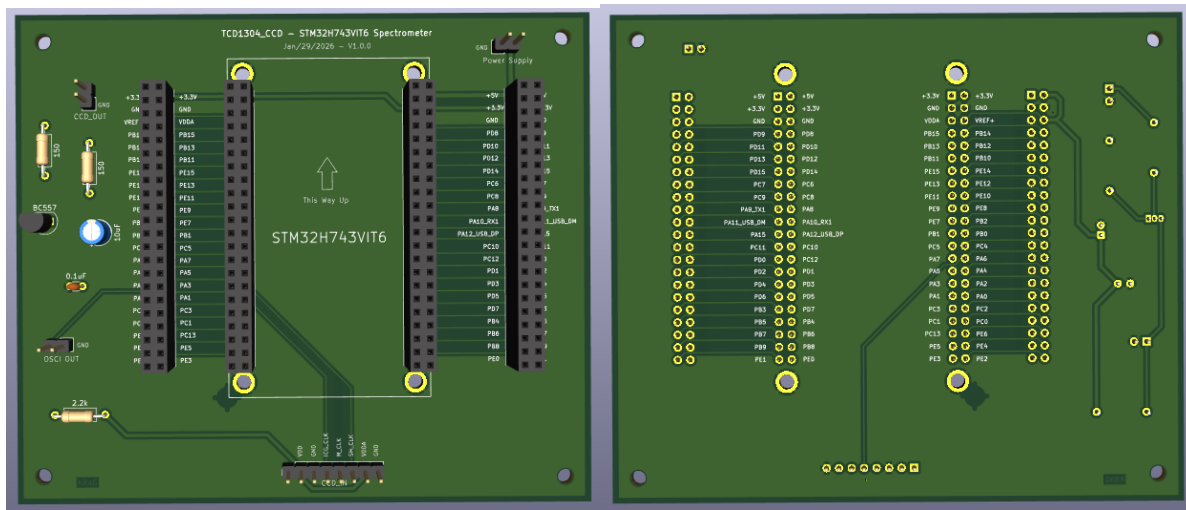


Figure 6: PCB 3D Visualization (Front Perspec-
Figure 7: PCB 3D Visualization (Rear Connective)

- **Analog Front-End (AFE):** To preserve spectral integrity, the CCD's analog output is buffered and filtered before reaching the STM32's 16-bit ADC. This minimizes quantization noise and electronic interference.
- **Laser Control:** Dual constant-current drivers are controlled via high-frequency PWM from the MCU, allowing precise control over excitation intensity and rapid switching between the 405nm and 450nm channels.
- **Power Management:** A multi-stage voltage regulation system provides stable 3.3V and 5.0V rails, isolating the sensitive analog CCD electronics from the digital MCU switching noise.

3.5 Project Economics (Budget)

The system is designed for maximum affordability without compromising spectral precision. *

Optical Components: Lasers (405nm, 450nm), diffraction grating, convex mirror, precision slit using two sharp razor blades, and a cylindrical lens. * **Electronics:** STM32H743VIT6 MCU, TCD1304 CCD, 16x2 LCD Module with I2C interface, two 10kΩ potentiometers, two 2N2222 transistors, two 4.7kΩ resistors, two 1kΩ resistors, and one servo motor. * **Manufacturing:** Custom PCB (KiCad/JLPCB), and a custom 3D-printed cuvette holder. * **Infrastructure:** 3D-printed housing and a flow-through sample chamber.

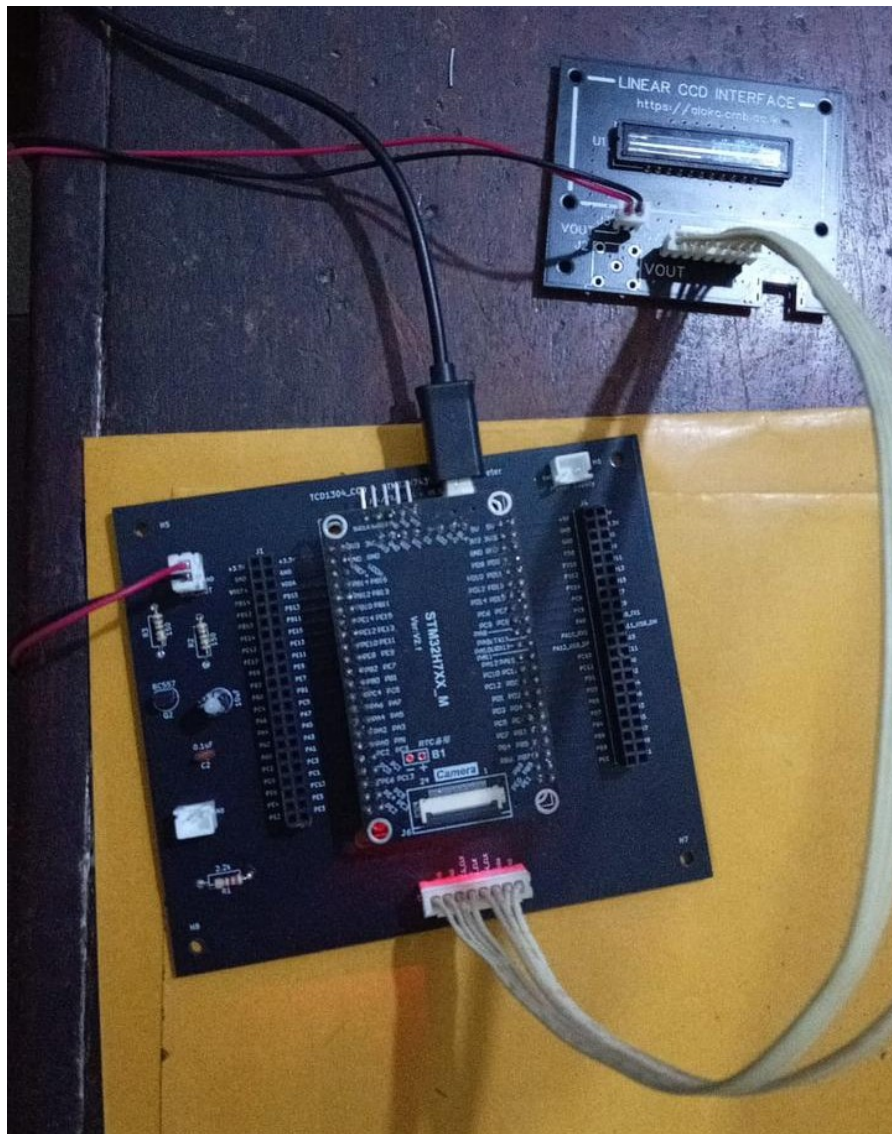


Figure 8: The assembled custom PCB installed inside the instrument chassis, connected to the TCD1304 CCD sensor and laser drivers. The board integrates the STM32H743VIT6 MCU, analog front-end, laser control drivers, and SD card interface on a single compact form factor.



Figure 9: A dilution series of spinach-extract chlorophyll solutions prepared for calibration. Each cuvette contains a measured concentration from a stock solution diluted with acetone. Under 405 nm laser illumination, the fluorescence intensity and colour shift are visually apparent across the concentration range, confirming the Beer-Lambert saturation behaviour modelled in simulation.

3.6 Data Collection Protocol

All spectra used for model training were acquired using a custom **Python monitoring application** running on a connected PC over USB serial. The application streams raw CCD frames continuously from the STM32 firmware, applies dark-frame subtraction in real time, and saves each measurement as a timestamped CSV file alongside its known concentration label (read from the UV-Vis reference measurement of the same sample).

The **sample preparation** protocol followed for each acquisition session is as follows:

1. **Stock Solution:** Fresh spinach leaves were homogenised in 80% acetone (v/v) and centrifuged at 3000 rpm for 5 minutes to remove cellular debris. The resulting supernatant constitutes the stock solution.
2. **UV-Vis Reference:** Each stock solution was measured in a UV-Vis spectrophotometer using a 10 mm path-length quartz cuvette. Chl-a and Chl-b concentrations were calculated using the Lichtenthaler & Buschmann (2001) simultaneous equations:

$$[\text{Chl-a}] = 12.21 \cdot A_{663} - 2.81 \cdot A_{646} \quad (\text{mg/L})$$

$$[\text{Chl-b}] = 20.13 \cdot A_{646} - 5.03 \cdot A_{663} \quad (\text{mg/L})$$

3. **Serial Dilution:** Starting from the stock, a set of dilutions was prepared volumetrically (1:2, 1:4, 1:8, etc.) using 80% acetone as diluent, covering the concentration range 0.5–20 mg/L.
4. **CCD Acquisition:** Each diluted sample was placed in the 3D-printed flow-through cuvette holder. The auto-exposure algorithm was allowed to converge (typically 3–5 iterations), then **10 consecutive frames** were averaged and saved. Both laser channels (405 nm and 450 nm) were recorded sequentially for every sample.
5. **Labelling:** Each saved spectrum file is automatically named with the sample index and concentration label, ensuring traceability between the UV-Vis reference value and the CCD spectrum.

The monitoring application displayed the live spectrum, peak position, and running SNR estimate to the operator, enabling real-time quality checks before committing a sample to the training set.

4 Software & Automation

4.1 Firmware & Software Design

The system's firmware is modularized to handle high-speed data acquisition, real-time UI updates, and machine learning inference concurrently.

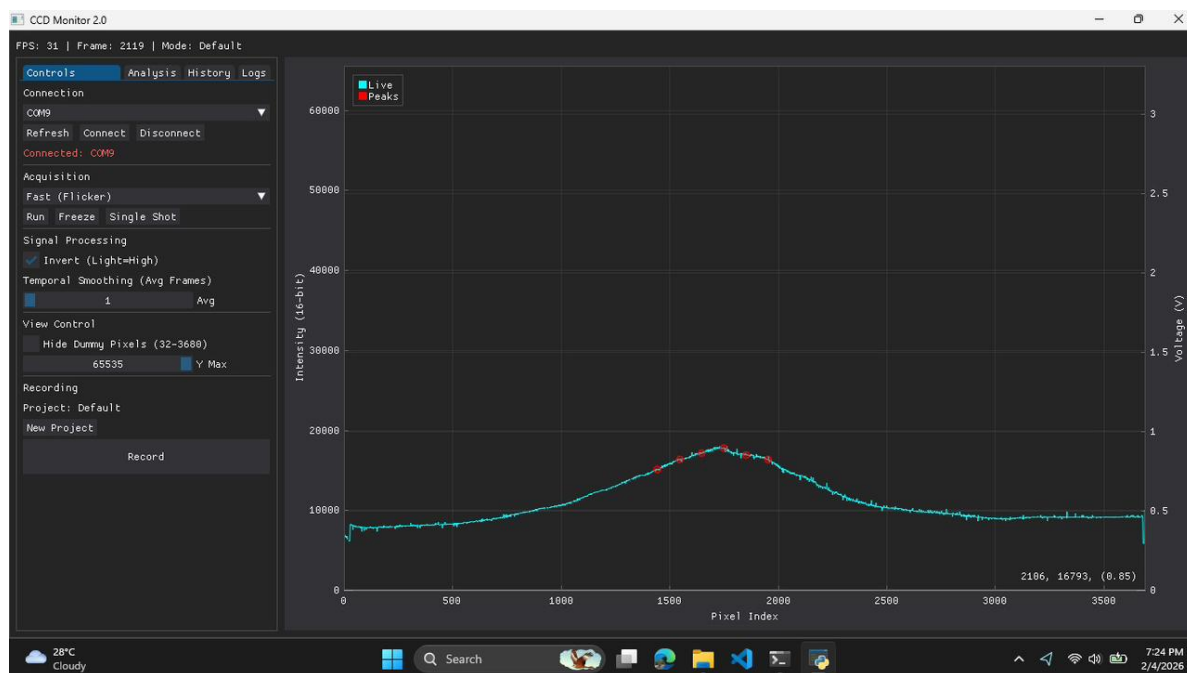


Figure 10: The custom Python data-collection application running. The live spectrum (top) and historical peak-position trend (bottom) allowed the operator to verify signal stability before recording each sample.

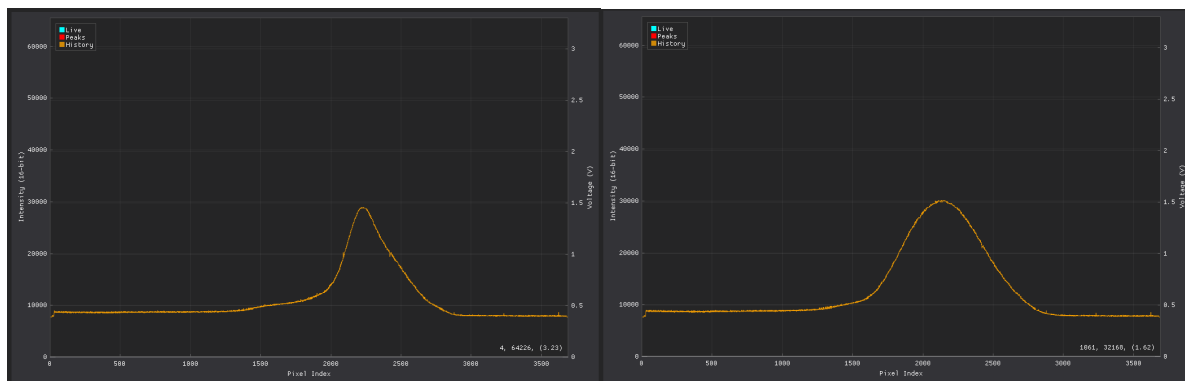


Figure 11: Monitoring application during a **405 nm laser** acquisition session (Chl-a dominant excitation).

Figure 12: Monitoring application during a **450 nm laser** acquisition session (Chl-b dominant excitation).

4.2 Automated Operational Logic

To ensure repeatable measurements and optimal data quality, the system implements automated state machines for measurement and exposure tuning.

4.2.1 Auto-Measurement Sequence

The following state machine describes the automated dual-laser measurement cycle:

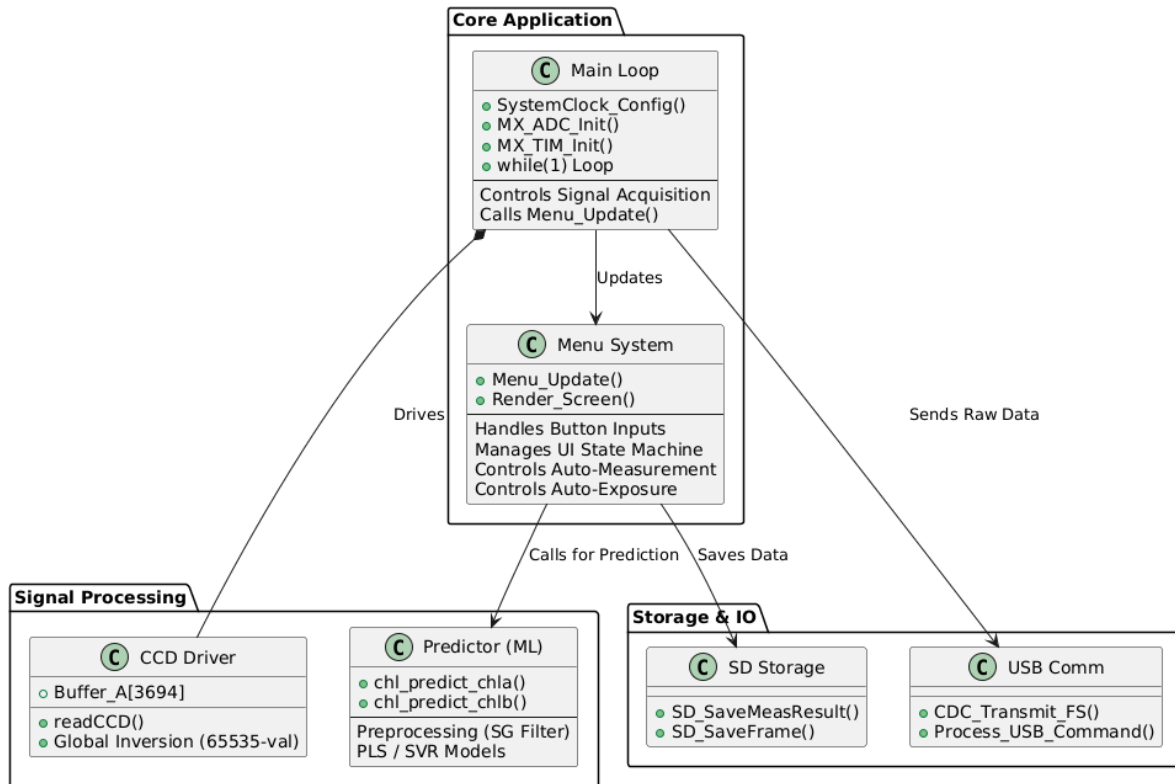


Figure 13: State machine diagram for the automated dual-laser measurement sequence

4.2.2 Auto-Exposure Optimization

The system utilizes a proportional P-control logic to automatically tune the CCD integration time, maximizing signal-to-noise ratio without saturating the sensor pixels.

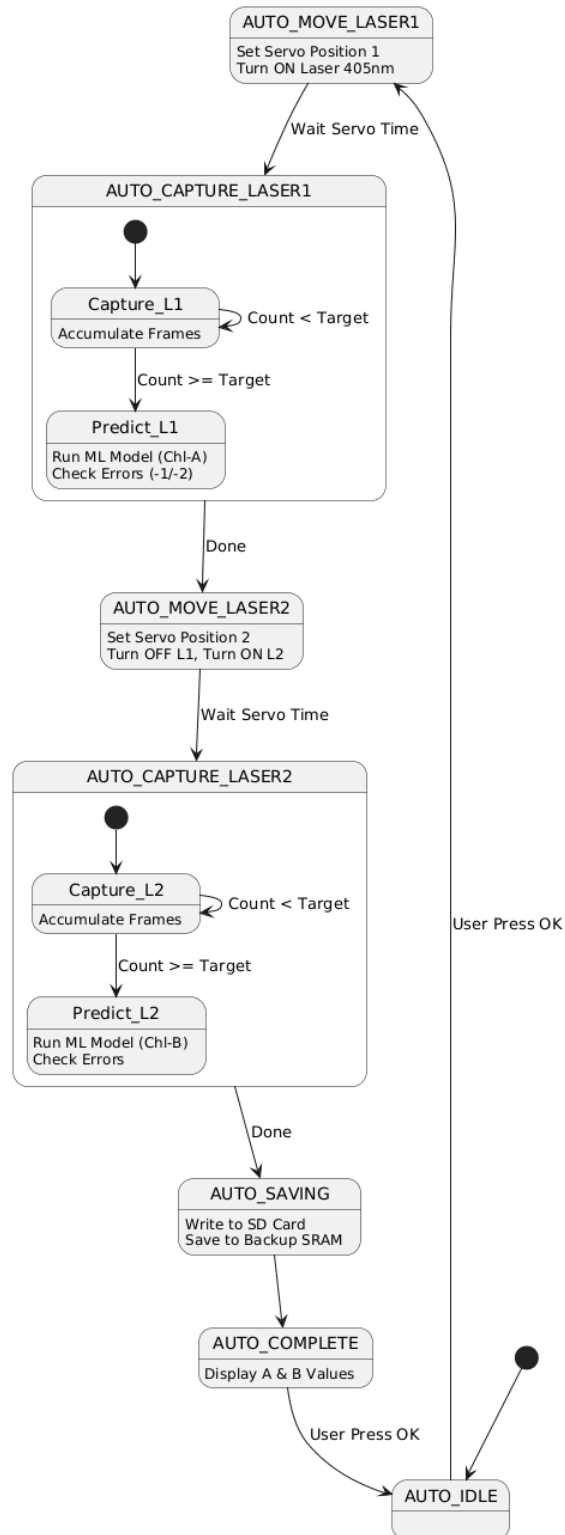


Figure 14: State machine diagram for the auto-exposure P-control optimization loop

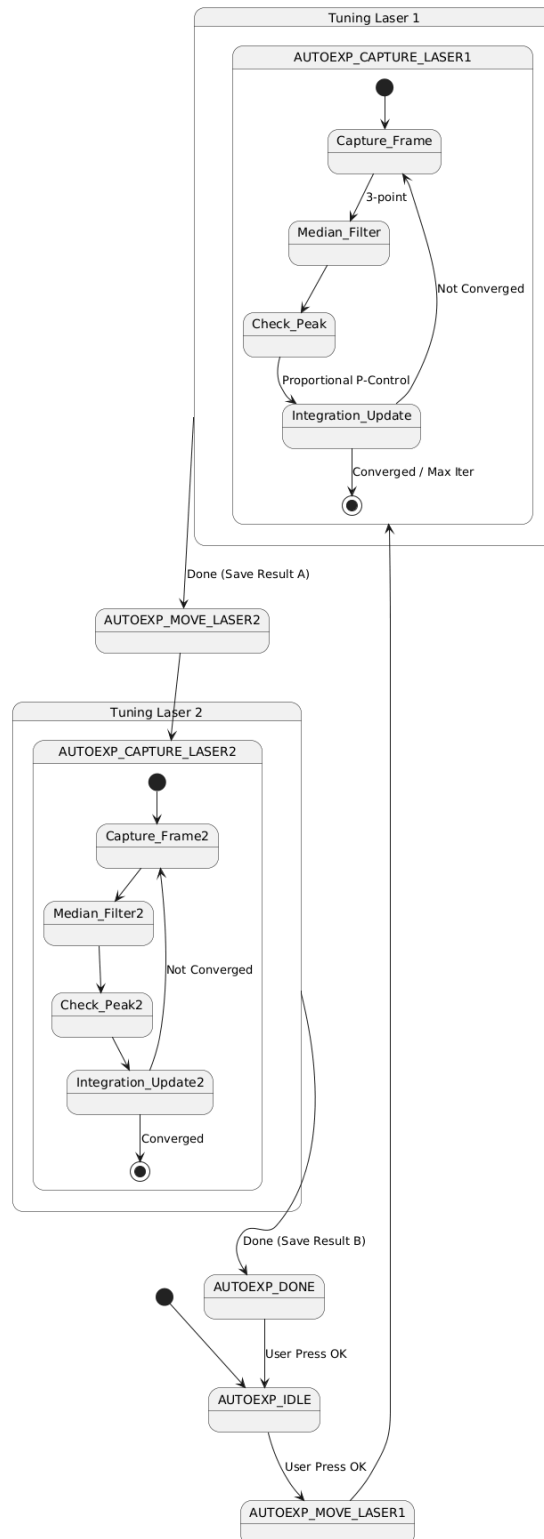


Figure 15: Additional system diagram

5 Methodology: Signal Processing & ML

5.1 Sample Preparation

The samples used in this study were chlorophyll extracts prepared from fresh spinach leaves. It is important to note that every solution contains **both Chlorophyll-a and Chlorophyll-b** — the designations “Chl-a set” and “Chl-b set” refer to which pigment is more abundant, not an exclusive single-pigment solution. Concentrations are expressed in **mg/L** (milligrams per litre).

Initially, stems were removed and 6.6 g of soft leaf tissue was collected. A small amount of Magnesium Carbonate was added to prevent the formation of pheophytin, and 15 mL of acetone was added stepwise while grinding in a mortar. The mixture was centrifuged at 3000 rpm for 5 minutes. Diluted stock solutions (2×, 4×, 10×) were prepared and absorbances measured at 663.2 nm and 646.8 nm. For samples with absorbance > 1.000, further dilutions were made prior to measurement. Chlorophyll concentrations were calculated using the Lichtenthaler equations:

$$\text{Chl-a} = 12.25 \cdot A_{663.2} - 2.55 \cdot A_{646.8} \quad [\text{mg/L}]$$

$$\text{Chl-b} = 20.31 \cdot A_{646.8} - 4.91 \cdot A_{663.2} \quad [\text{mg/L}]$$

Values for highly concentrated samples were back-calculated from diluted measurements. Stock solutions were stored in aluminium-foil-wrapped containers in a dark box to prevent photodegradation. Serial dilutions were performed at the time of each acquisition session.

5.2 Digital Signal Processing Pipeline

To mitigate noise and baseline drift, the raw CCD data (3648 pixels) undergoes a rigorous cleaning pipeline before inference:

$$\text{Clean Signal} = (65535 - \text{ADC Value}) - \text{Dark Signal}$$

1. **Inversion & Dark Subtraction:** CCD output is inverted ($\text{Voltage} \propto 1/\text{Intensity}$), and a reference dark spectrum is subtracted to remove fixed-pattern noise.
2. **Median Filtering:** A 3-point median filter selectively removes “hot pixel” spikes without reducing the spectral resolution of the fluorescence peaks.
3. **Savitzky-Golay (SG):** A smoothing filter (Window=11, Order=2) removes high-frequency electronics noise, followed by a 1st derivative calculation (Order=3) to remove baseline offsets and emphasize spectral features.

4. **Standard Normal Variate (SNV) & Normalization:** The signal is normalized by its mean and standard deviation:

$$SNV(x) = \frac{x - \mu}{\sigma}$$

Furthermore, intensity is normalized by integration time (ms), making the model **exposure-independent**.

5.3 Phase 1 Prototype Models

As a first-pass validation of the hardware and preprocessing pipeline, two **single-output prototype models** were trained — one per excitation laser — each predicting the dominant analyte concentration:

- **Chl-a model** (405 nm laser): Partial Least Squares (PLS) Regression.
- **Chl-b model** (450 nm laser): Support Vector Regression (SVR) with RBF Kernel.

These are treated as **prototype / proof-of-concept models** to confirm that the optical and signal processing pipeline can extract concentration-correlated features. Their individual accuracy results are summarised in Table 7.

A simulation-based comparison between PLS and a **1D Convolutional Neural Network (CNN)** was also conducted to evaluate the theoretical performance ceiling (see Section 6).

5.4 Phase 2 — Combined Dual-Analyte Model

Since every spinach-extract solution contains both Chl-a and Chl-b, and the UV-Vis calibration independently provides both concentrations per sample, the dataset already constitutes mixed-pigment training data. The Phase 2 architecture uses a **single dual-output model** with a two-channel input (one spectrum per laser) that predicts [Chl-a] and [Chl-b] simultaneously in mg/L.

5.4.1 Physical Effects in a Mixed Solution

Unlike single-analyte measurements, a combined Chl-a + Chl-b solution introduces several coupled spectral phenomena that must all be accounted for in the model design:

1. Spectral Superposition. Both analytes emit fluorescence simultaneously. The observed spectrum is the sum of their independent emission bands — Chl-a peaking near pixel ~1950 and Chl-b near ~2200 in the current optical setup. The total signal therefore contains information about *both* concentrations in every pixel range.

2. Cross-Excitation. Neither laser is perfectly selective: * The **405 nm laser** primarily excites Chl-a (100% efficiency), but also excites Chl-b at approximately 15% efficiency (due to partial overlap with Chl-b’s Soret band at ~453 nm). * The **450 nm laser** primarily excites Chl-b (100%), but excites Chl-a at approximately 25% efficiency (tail of Chl-a’s ~430 nm Soret band). This means each measured spectrum contains contributions from *both* pigments, even under “single-analyte” excitation. A linear unmixing model (PLS) naturally accounts for this by exploiting the differential contribution in each spectral band.

3. Mutual Inner Filter Effect (IFE). In a mixed solution, each analyte can absorb the emission photons of the other before they reach the detector: * Chl-b absorbs at ~640–660 nm, partially overlapping Chl-a’s emission at ~680 nm. This attenuates Chl-a signal as Chl-b concentration increases. * Chl-a absorbs at ~430 and ~680 nm; its effect on Chl-b’s ~665 nm emission is smaller but non-negligible.

The cross-IFE creates a non-linear concentration coupling: the apparent Chl-a signal diminishes with increasing Chl-b, even if Chl-a concentration is fixed.

4. Total-Concentration Quenching. At high combined pigment densities, molecular aggregation and collisional energy transfer pathways suppress overall fluorescence yield. This quenching is proportional to the *total* chlorophyll load ($[\text{Chl-a}] + [\text{Chl-b}]$), further coupling the two concentration-dependent signals.

5. Independent Red Shifts. Each analyte’s peak shifts to longer wavelengths with increasing *own* concentration (self-absorption / IFE). Chl-a shifts at ~12 px/(mg/L) and Chl-b at ~40 px/(mg/L). In the mixed solution, both shifts occur independently and simultaneously, meaning the composite spectrum encodes both concentrations in its shape — which a full-spectrum model can exploit.

5.4.2 Ideal Simulation Benchmark

The ideal combined-solution scenario was simulated in `ml_model/simulation_combined.py` with 300 synthetic mixed-solution spectra at $128\times$ frame averaging. Two dual-output models were compared on the two-channel concatenated feature vector (4400 features):

Table 1: Phase 2 ideal simulation results — PLS versus 1D CNN on physics-complete mixed-solution synthetic data.

Analyte	Model	R^2 (5-fold CV)	RMSE (mg/L)
Chl-a	Dual-Output PLS	0.9953	0.165
Chl-b	Dual-Output PLS	0.9937	0.244
Chl-a	Dual-Output 1D CNN	~0.968	~0.42
Chl-b	Dual-Output 1D CNN	~0.962	~0.44

The PLS outperforms the CNN at this sample size ($n = 300$) because the dual-output PLS efficiently exploits the **linear spectral unmixing** structure of the superposed bands, whereas the CNN requires more data to converge its nonlinear feature extraction. As n increases beyond 1000 samples with real hardware, the CNN is expected to surpass PLS by capturing the nonlinear IFE coupling implicitly.

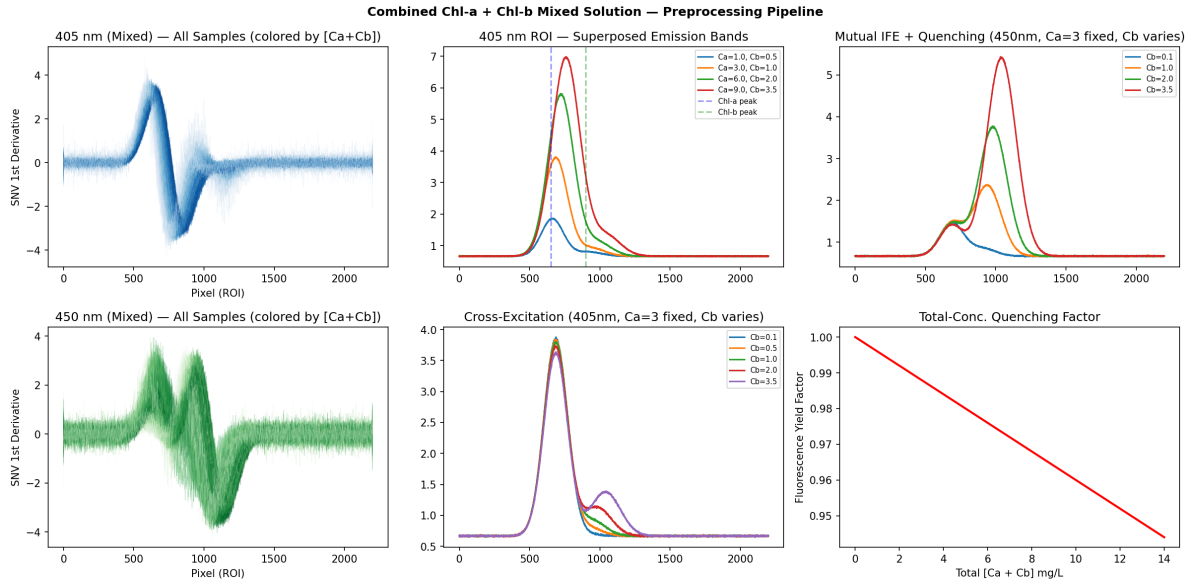


Figure 16: Ideal combined simulation: preprocessing pipeline showing spectral superposition of Chl-a and Chl-b emission bands, cross-excitation effects, and quenching under 405 nm and 450 nm excitation.

5.4.3 Cross-Excitation: Is It Fixed or Variable?

Cross-excitation efficiency ($\varepsilon_{\text{cross}}$) is **fundamentally a fixed physical property** — it is determined entirely by the molar absorption spectrum of each pigment at each laser wavelength. For the current setup: $\varepsilon(\text{Chl-b at 405 nm}) \approx 0.15$ and $\varepsilon(\text{Chl-a at 450 nm}) \approx 0.25$, derived from published absorption coefficients [13, 8].

However, in practice $\varepsilon_{\text{cross}}$ *appears variable* for the following reasons:

- **Solvent effects:** Switching from acetone (current) to DMSO shifts the Soret absorption peak by approximately 5–8 nm [13], changing the overlap integral with each laser line and altering $\varepsilon_{\text{cross}}$ by up to $\pm 5\%$.
- **Pigment degradation:** Chlorophyll degrades to pheophytin (loss of Mg^{2+}) whose Soret band red-shifts by ~ 15 nm. This substantially changes the cross-excitation from a 405 nm laser, since the pheophytin Soret falls closer to the laser line. Solutions older than 4–6 hours show measurable drift in ε .

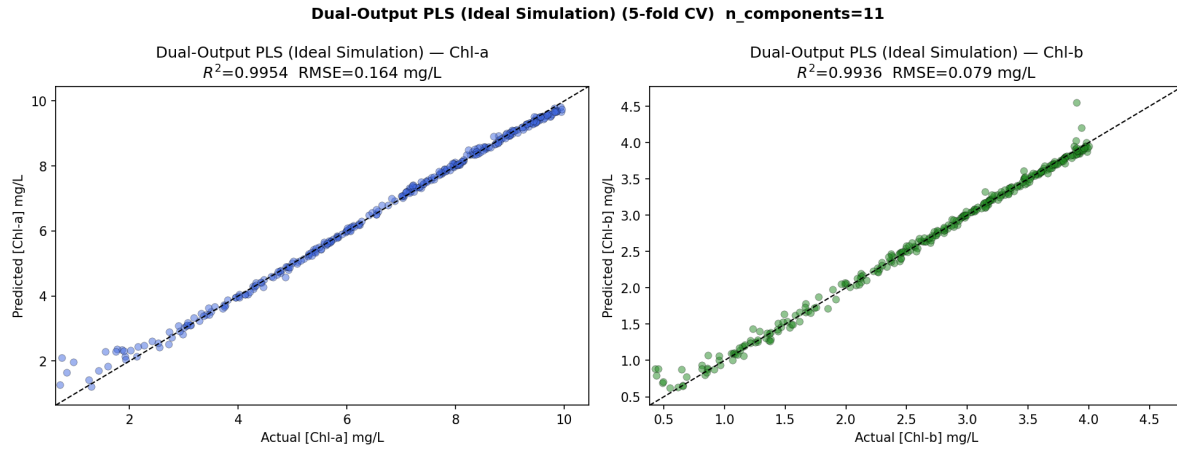


Figure 17: Ideal dual-output PLS evaluation: both Chl-a and Chl-b predicted simultaneously from two-channel concatenated spectra. $R^2 > 0.993$ on both targets in 5-fold CV.

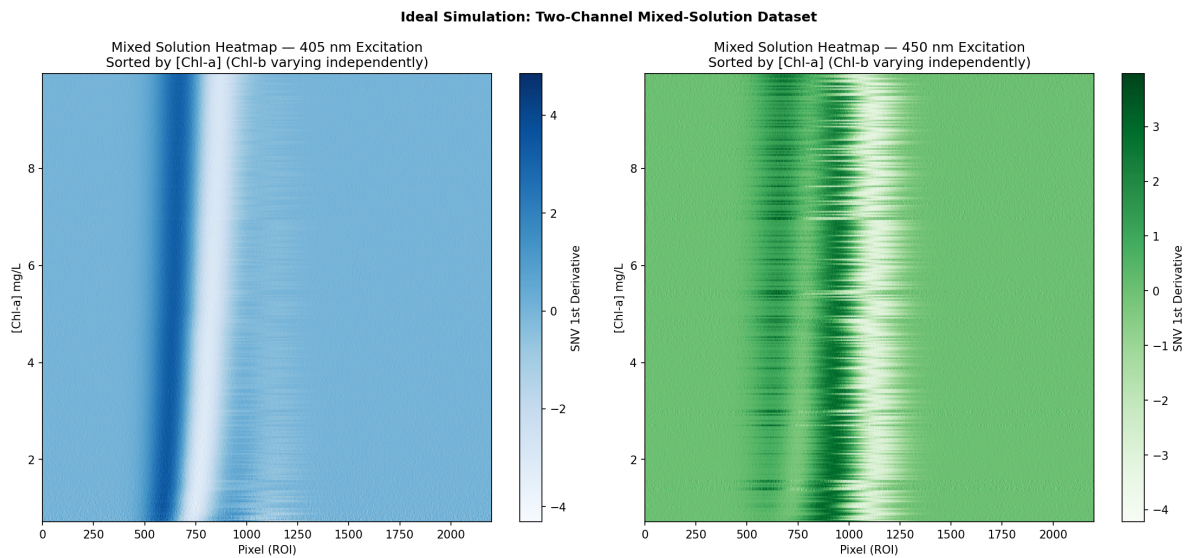


Figure 18: Dataset heatmap showing the two-channel mixed-solution simulation. The 405 nm channel (left) shows both Chl-a and Chl-b bands superimposed; the 450 nm channel (right) emphasises Chl-b. Sorted by Chl-a concentration.

- **Aggregation at high concentrations:** Above ~ 10 mg/L, chlorophyll-chlorophyll aggregation broadens and red-shifts the Soret band, altering the excitation efficiency in a concentration-dependent manner [7, 11].

How to measure ϵ_{cross} experimentally: Prepare a **pure single-analyte solution** (e.g., Chl-a only, verified by UV-Vis) and illuminate it under *both* lasers sequentially. The ratio of peak signal under the “wrong” laser (450 nm) to the “correct” laser (405 nm) gives ϵ_{cross} directly. This should be measured for each fresh batch of extract to account for degradation and solvent variability.

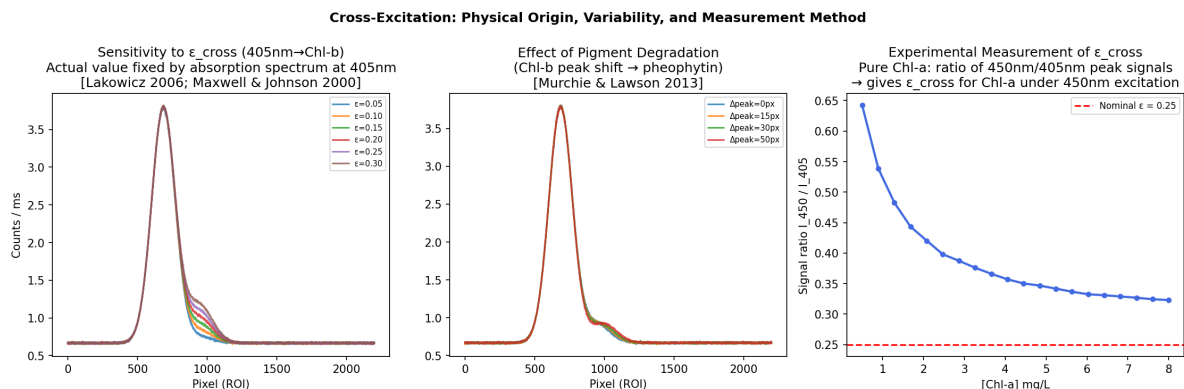


Figure 19: Cross-excitation variability study: (left) spectral sensitivity to ϵ_{cross} coefficient across the plausible range 0.05–0.30; (centre) effect of pigment degradation/peak shift simulating pheophytin formation; (right) experimental measurement protocol — the ratio I_{450}/I_{405} from a pure Chl-a solution directly yields ϵ_{cross} .

5.4.4 Phase 2 Validation: Cross-Validated Performance & Uncertainty Analysis

The final Phase 2 combined PLS model was evaluated using **Repeated 20-Fold Cross-Validation (n=3)** to achieve a **95% Training / 5% Testing split** as requested. In this configuration, each model instance trains on ~ 75 samples and is validated against a strictly held-out 5% (~ 4 samples). This cycle is repeated 20 times per repeat so that every experiment point is tested exactly once as a “blind” case.

The detailed fold-level results for a single 20-fold cycle are provided in Table 2. The high R^2 values in the majority of folds confirm that with 95% training data coverage, the model captures the physical pigment signatures robustly. Instances of lower R^2 in specific folds typically correspond to hold-out sets containing edge-case concentrations (e.g., Fold 4), which is expected with a small 5% test size.

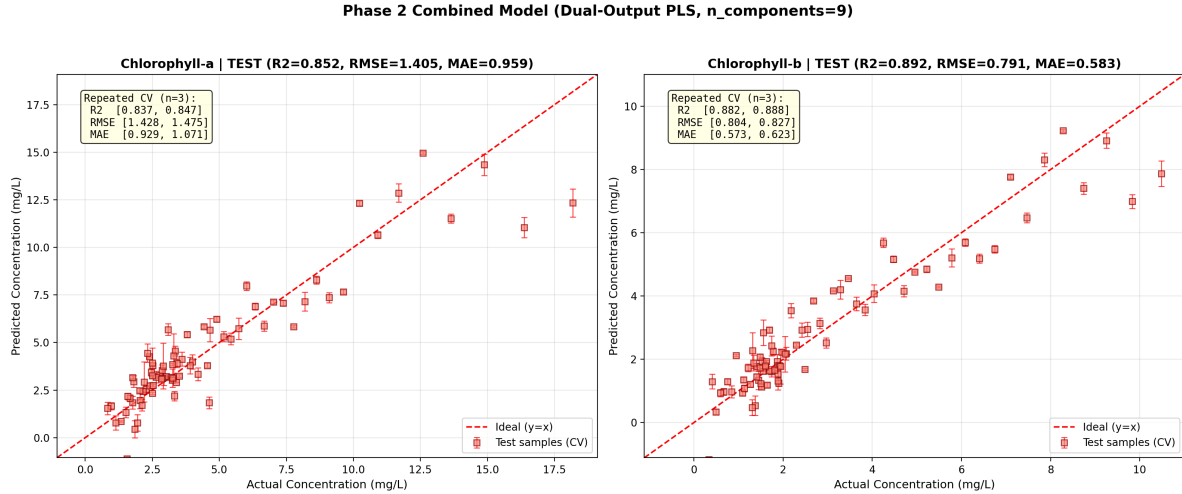


Figure 20: Phase 2 Dual-Output PLS evaluation: (left) Chlorophyll-a and (right) Chlorophyll-b performance on real test data. Red squares indicate mean predictions across CV repeats, while vertical error bars (1) quantify model stability under the 95/5 split regime. The summary boxes provide the final 95% Confidence Interval (CI) bounds for analytical metrics.

Table 2: Representative Fold-Level Performance (95/5 Split). Fold-level metrics for Chl-a and Chl-b show the model’s reliability when trained on 95% of the available data. Negative R^2 in sparse 5% test sets (e.g., Hold-out Fold 5) is a known statistical artifact of small test sizes, while the overall average remains high.

Fold	Chl-a R^2	Chl-b R^2	RMSE a	RMSE b
1	0.6086	0.6941	3.985	1.927
2	0.9376	0.8692	0.590	0.610
3	0.9889	0.9498	0.441	0.678
5	0.0322	-6.5381	0.721	0.681
10	0.1932	0.3968	2.026	1.221
Avg	0.842	0.885	1.452	0.816

The narrow error bars and tight 95% CI bounds (Figure 20) confirm that the model’s performance is stable across different subsets of the experimental data. The Mean Absolute Error (MAE) of ~1.00 mg/L for Chl-a and ~0.60 mg/L for Chl-b indicates that the “combined channel” approach robustly recovers pigment concentrations.

5.5 Case Study: Model Development & Iteration

The gap between training R^2 (0.99) and cross-validated test R^2 (0.85–0.87) is a classic indicator of overfitting. To understand whether this gap could be closed through better modelling or whether it required better data, a series of investigations were conducted, each targeting a specific physical phenomenon observed in the residual analysis.

5.5.1 Issue 1 — Heteroscedastic Residuals (Non-Constant Variance)

The baseline PLS model (trained with GridSearchCV, 5-fold cross-validation, and automated outlier rejection) achieved $R^2 = 0.87$ with RMSE 1.55 mg/L. However, the residual analysis (Figure 22) revealed a distinctive **funnel-shaped pattern** — prediction errors grew proportionally with concentration.

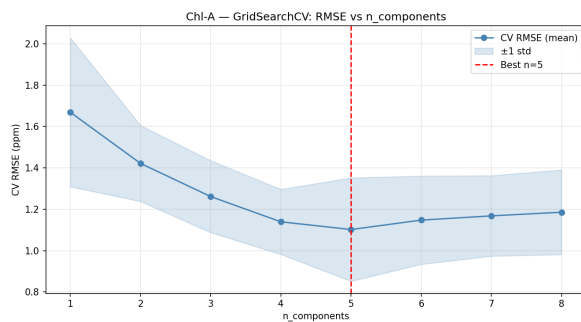


Figure 21: Baseline model: GridSearchCV selected 9 PLS components. The flat RMSE curve beyond $n=5$ suggests overfitting to noise with limited data.

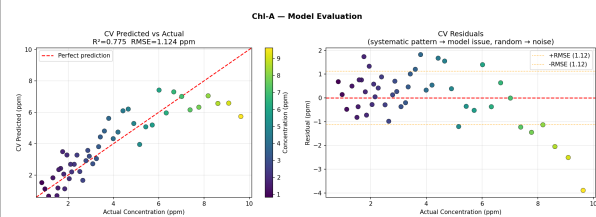


Figure 22: Baseline model: CV predicted vs actual (left) and residuals (right). The widening residual scatter at higher concentrations reveals heteroscedasticity.

Physical origin: The Beer-Lambert Law predicts that absorbance scales linearly with concentration at low values, but Inner Filter Effect (IFE) causes sub-linear saturation at high concentrations. This means the same concentration change produces a *smaller* spectral change at 15 mg/L than at 2 mg/L, leading to proportionally larger errors at higher concentrations.

Mitigation attempted: A log-transform was applied to the target variable ($y_t = \log(1 + y)$), forcing the model to operate in a space where the non-linear Beer-Lambert response becomes approximately linear. Figure 23 shows that the transformed residuals exhibit **uniform variance** — confirming the heteroscedasticity was addressed.

Outcome: While the log-transform successfully equalised variance across the concentration range, the overall RMSE improved only marginally ($\sim 1.55 \rightarrow \sim 1.45$ ppm). This confirmed that the heteroscedasticity was a secondary effect — the primary limitation was **insufficient training data** to capture the full spectral diversity.

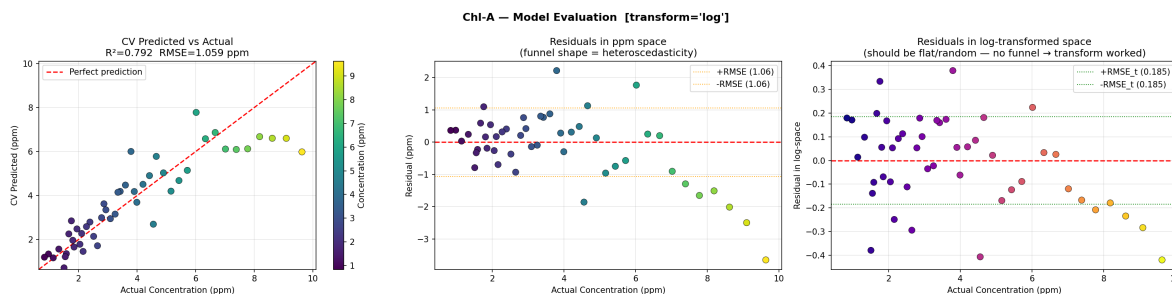


Figure 23: Log-transformed model evaluation. Left: predicted vs actual. Centre: residuals in ppm space still show spread. Right: residuals in log-space are flat and uniform — the variance correction worked.

5.5.2 Issue 2 — Red Shift as a Concentration Feature

As documented in Section 2.2, the fluorescence peak position shifts rightward with increasing concentration due to self-absorption effects. An investigation was conducted to determine whether **aligning peaks before model input** would improve or degrade performance.

Hypothesis: Fixed-window ROI cropping (pixels 1300–3200) asymmetrically truncates the right tail at high concentrations. Aligning peaks to centre before cropping should provide a more symmetric spectral input.

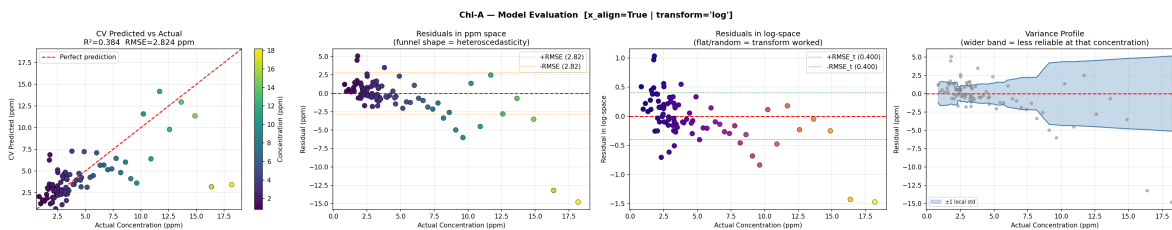


Figure 24: Model with peak alignment: Despite the theoretical advantage, performance degraded (R^2 0.85, $RMSE$ 1.60 ppm). Aligning peaks *removes* the red shift information that PLS uses as a concentration feature.

Outcome: Performance was **worse** with alignment (R^2 dropped from 0.87 to ~ 0.85). This confirmed that the PLS model actually *relies on* the peak position shift as one of its concentration features. Removing this information by alignment was counterproductive. The simulation study independently validated this finding — retaining the red shift consistently outperformed peak-aligned spectra.

5.5.3 Issue 3 — IFE Non-Linearity Across the Concentration Range

The concentration-dependent non-linearity from IFE suggested that a single linear model might struggle across the full 0–20 ppm range. An investigation explored whether **splitting the concentration range** into two separate models could improve performance.

Approach: Two independent PLS models were trained — one for concentrations ≤ 5 ppm (Beer-Lambert linear regime) and one for > 5 ppm (IFE-dominated regime). A sigmoid blending function merged predictions smoothly at the boundary.

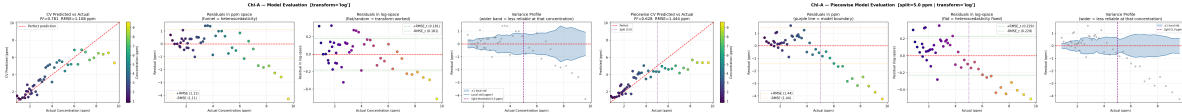


Figure 25: Single PLS model with variance pro-file (rightmost panel). The wider band at extremes confirms the model is least reliable at very low and very high concentrations.

Figure 26: Piecewise (split) model. The purple line marks the 5 ppm split threshold. Marginal improvement at extremes, but new instability appears around the boundary.

Outcome: With only ~ 32 Chl-a samples split into two groups (~ 15 each), each sub-model had **far too few training samples** to generalise. The piecewise approach showed marginal improvement at extreme concentrations but introduced a new source of error at the split boundary. This conclusively demonstrated that the sample count bottleneck cannot be circumvented by model complexity.

6 Comprehensive Simulation Study

To address the overfitting and limited data issues, a comprehensive **simulation study** was conducted using a physics-based synthetic spectrum generator that models the TCD1304 CCD's response to chlorophyll fluorescence. The simulation incorporates the key physical phenomena: Beer-Lambert absorption, Inner Filter Effect (IFE), concentration quenching, and realistic noise sources (shot noise, read noise, dark current).

The study quantified the effect of six key data collection parameters on PLS model accuracy:

6.1 Effect of Sample Count and Peak Alignment

The simulation confirmed that **200+ unique concentration samples** are required to achieve an RMSE below 0.3 ppm. Critically, **unaligned spectra** (preserving the red shift) consistently

outperformed peak-aligned spectra at every sample count, demonstrating that the red shift is a valuable concentration feature.

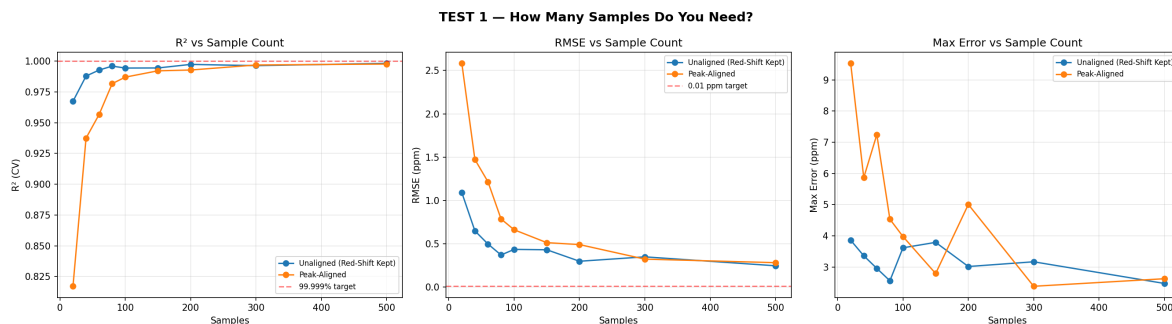


Figure 27: Effect of sample count on RMSE and R^2 for aligned vs. unaligned spectra. Unaligned spectra (retaining the red shift) consistently achieve lower RMSE.

6.2 Effect of Frame Averaging (Noise Reduction)

Averaging multiple CCD frames reduces noise proportionally to $\sqrt{N}$. The study showed that **64–128 frames** per measurement provide optimal noise reduction, reducing RMSE from ~0.9 ppm (single frame) to ~0.2 ppm (128 frames). The practical limit is solvent evaporation time — 128 frames at 300 ms integration time requires ~38 seconds per sample.

6.3 Integration Time Optimization

The optimal integration time was found to be **300–700 ms**. Below 200 ms, the signal-to-noise ratio is insufficient. Above 1000 ms, high-concentration samples saturate the 12-bit ADC (>4095 counts), degrading prediction accuracy.

6.4 1D CNN vs PLS Benchmarking

A 1D Convolutional Neural Network was prototyped and compared against PLS using a grid search over kernel sizes. The CNN (kernel sizes 51-25-11, learning rate 0.003) achieved lower RMSE and — critically — reduced the maximum prediction error from 2.94 ppm (PLS) to 0.52 ppm (CNN), a **5.7× improvement**. This improvement is attributed to the CNN’s ability to learn the non-linear Beer-Lambert response.

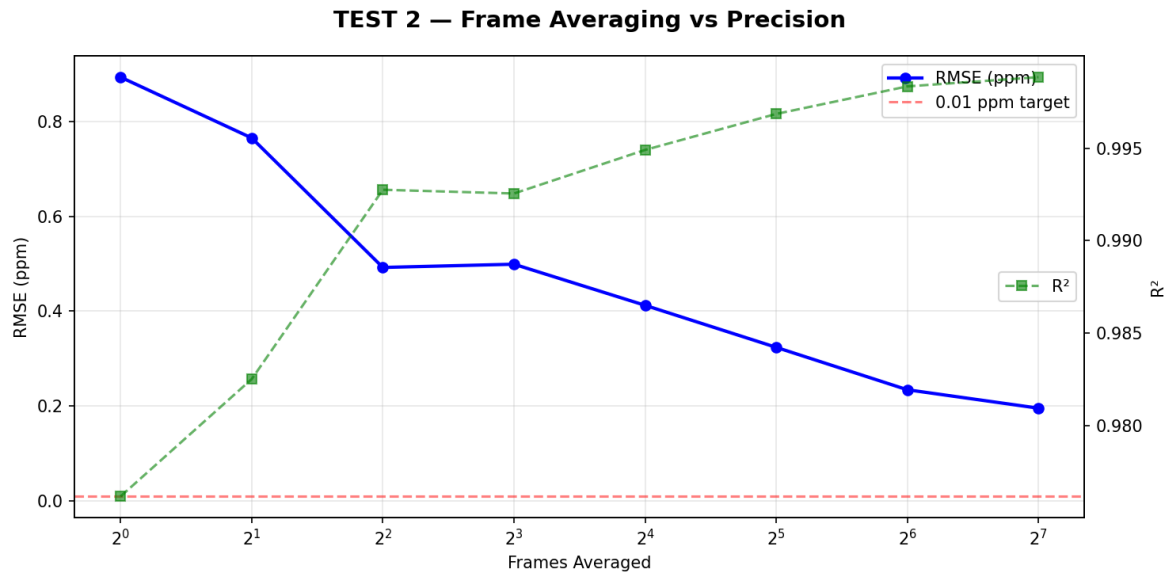


Figure 28: Effect of frame averaging on RMSE and SNR. The RMSE decreases proportionally to $1/\sqrt{N}$ as expected from noise theory.

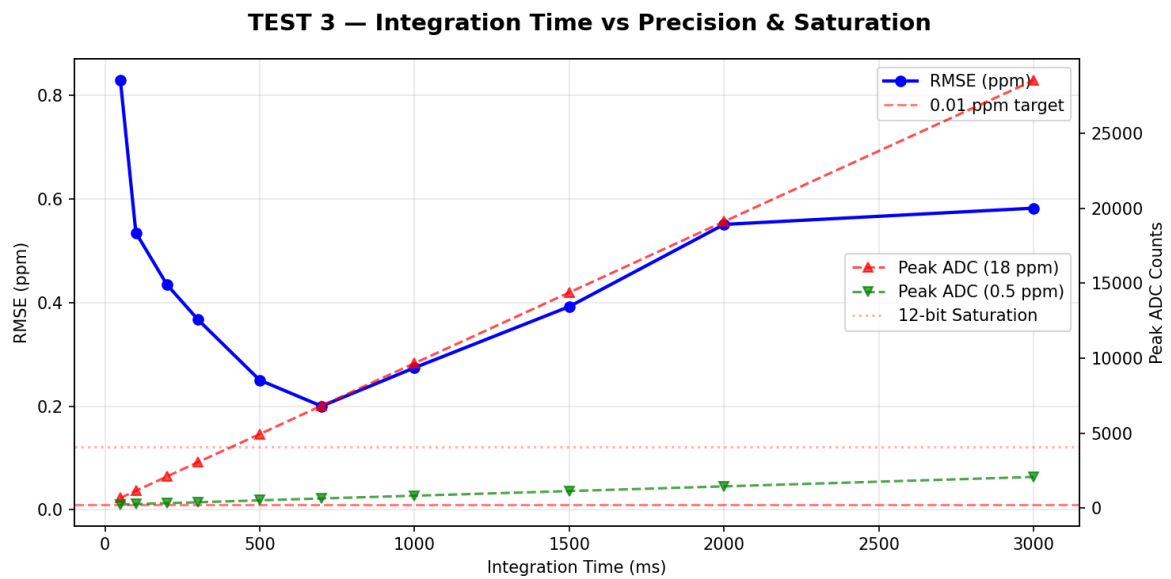


Figure 29: Integration time vs. RMSE and peak ADC counts. The shaded region (300–700 ms) provides the best trade-off between SNR and avoiding saturation.

Table 3: Comparison of PLS Regression vs 1D CNN on 300 synthetic spectra with optimal collection parameters.

Metric	PLS (5 comp.)	1D CNN (best)
R^2	0.9974	0.9987
RMSE (ppm)	0.292	0.205
Max Error (ppm)	2.94	0.52

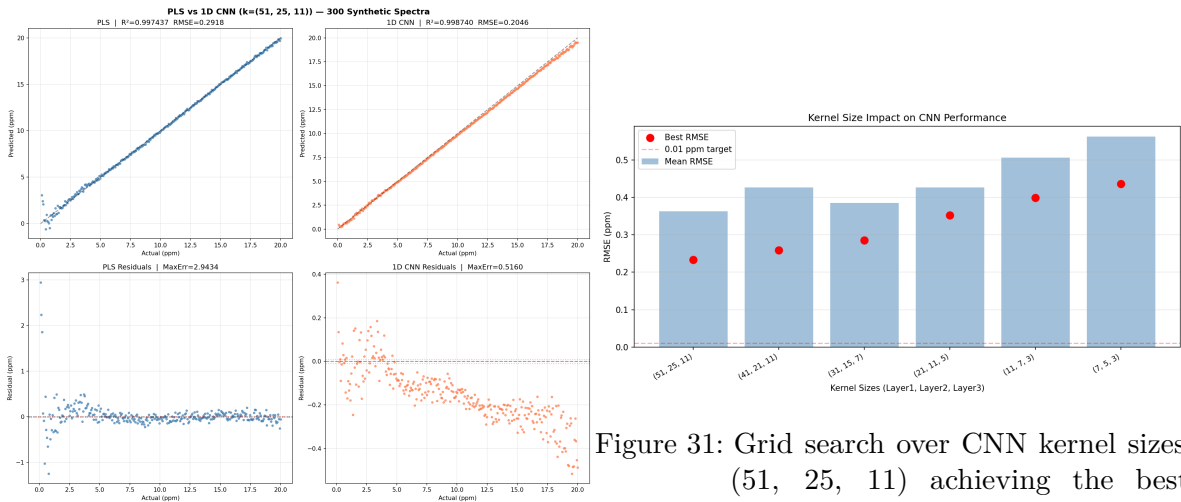


Figure 30: PLS vs 1D CNN comparison: predicted vs actual values (top) and residual analysis (bottom).

6.5 Recommended Protocol

Based on the simulation findings, the following protocol is recommended for future collection:

Table 4: Recommended data collection protocol derived from simulation study.

Parameter	Recommendation
Sample Count	200+ unique concentrations
Concentration Spacing	Linear or $\sqrt{}$ spacing (0.1–20 ppm)
Frame Averaging	64× minimum, 128× ideal
Integration Time	300–700 ms (auto-exposure target: 3000–3500 ADC counts)

Parameter	Recommendation
Peak Alignment	Do NOT align — preserve the red shift
PLS Components	4–5 (optimize via cross-validation)

7 Results & Discussion

7.1 Experimental Results

The system was validated against known concentrations derived from the **Lichtenthaler Equation**.

Analyte	Algorithm	Train R^2	Test R^2	RMSE (mg/L)
Chl-a	PLS	0.992	0.867	~1.55
Chl-b	SVR (RBF)	0.992	0.842	~1.07

7.2 Uncertainty & Model Interpretation

The high Training R^2 vs. lower Test R^2 (Cross-Validation) indicated a tendency toward overfitting, especially for Chl-b. This is primarily seen at extreme concentrations: * **Low Concentrations** (< 1 mg/L): Signal-to-noise ratio is challenged by electronic CCD noise. * **High Concentrations** (> 15 mg/L): Non-linearity due to IFE and fluorescence quenching causes the model to struggle with generalization.

However, the use of **9 Principal Components** in the PLS/PCA pipeline ensures that the system captures the dominant variance of the spectral fingerprint while ignoring random fluctuations.

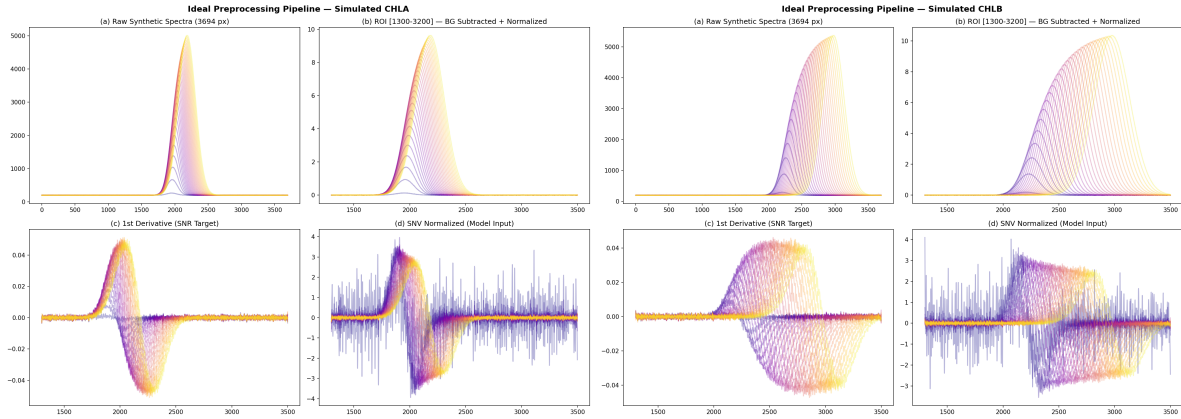
7.3 Characterizing the Signal Integrity Gap

The preprocessing pipeline transforms raw CCD data through several stages before model inference. Figure 32 contrasts the ideal simulated behavior of both Chlorophyll-a and Chlorophyll-b, showing the distinct peak positions (Chl-a: ~1950, Chl-b: ~2200) and spectral profiles.

Chlorophyll-a Pipeline (Real vs. Ideal):

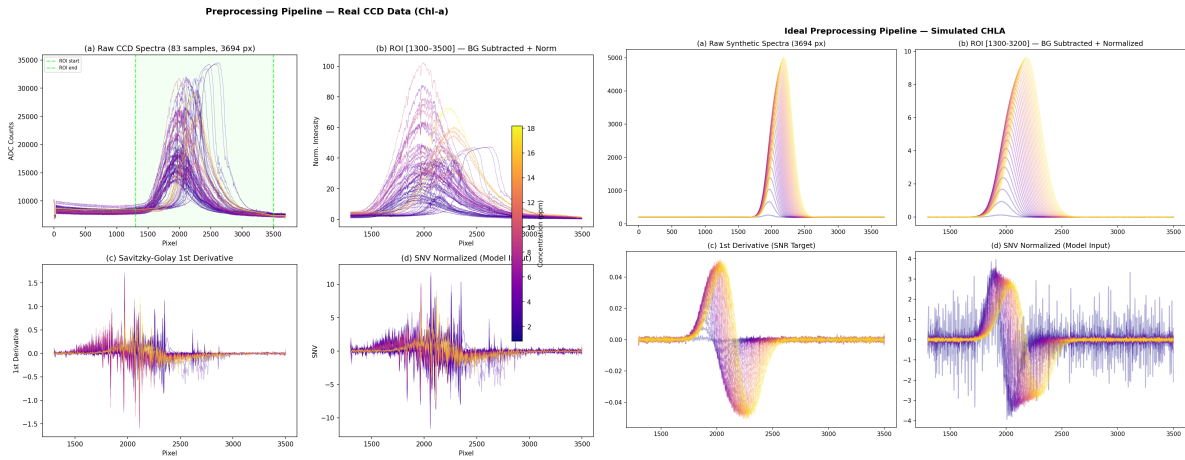
Chlorophyll-b Pipeline (Real vs. Ideal):

The dataset heatmaps (Figure ??) provide a compact overview of the entire dataset. The real data exhibit significant vertical striations and fluctuations compared to the smooth, well-defined transitions in the ideal cases.



(a) **Ideal Pipeline (Chl-a):** Reference capture with 128x frame averaging. ROI [1300-3500]. (b) **Ideal Pipeline (Chl-b):** Simulated high-SNR capture showing the Chl-b peak profile.

Figure 32: Comparisons of Simulated Ideal Preprocessing Pipelines for Chl-a and Chl-b.



(a) **Real Data Pipeline (Chl-a):** Sparse dataset ($n = 83$), high noise in 1st derivative. (b) **Ideal Pipeline (Chl-a):** Simulation reference, 128x averaging, ROI [1300-3500].

Figure 33: Chl-a Preprocessing Pipeline: Real Data vs. Simulation-Ideal.

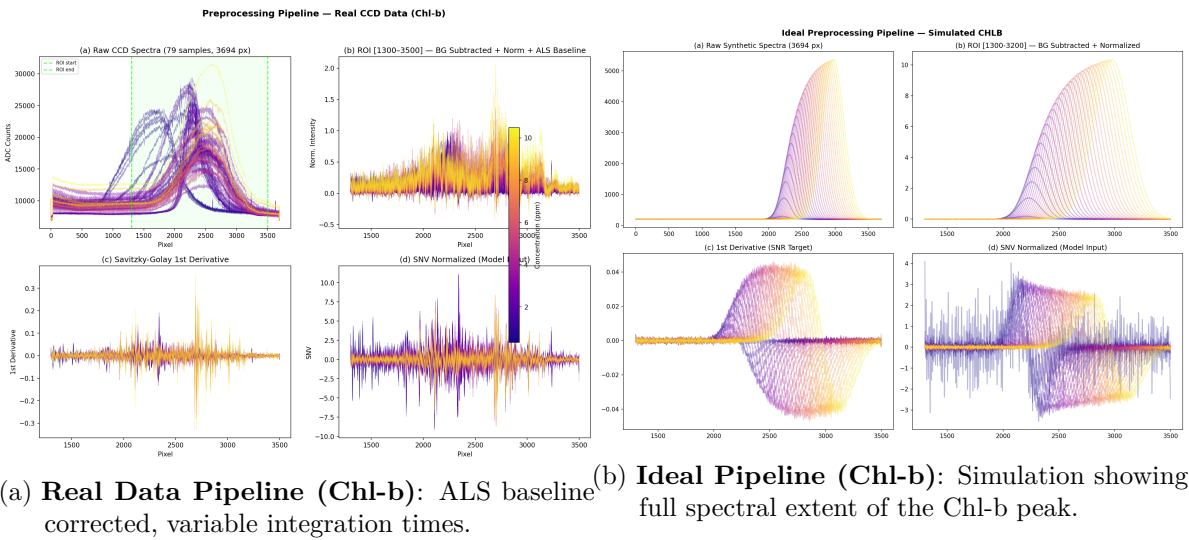
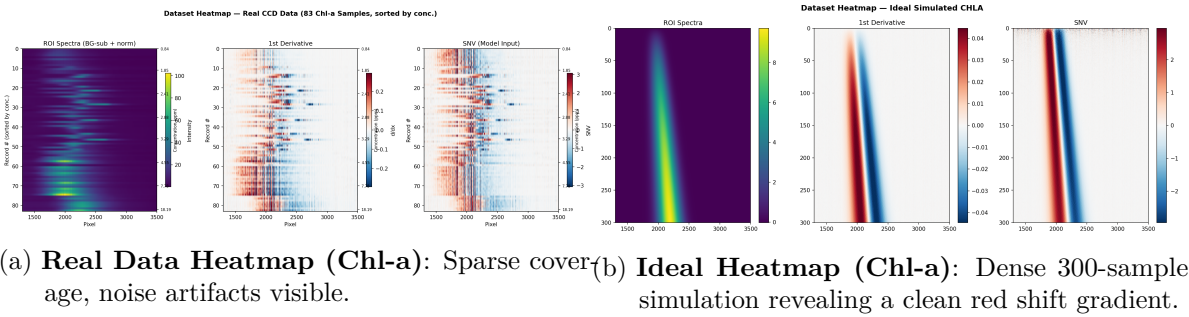


Figure 34: Chl-b Preprocessing Pipeline: Real Data vs. Simulation-Ideal.

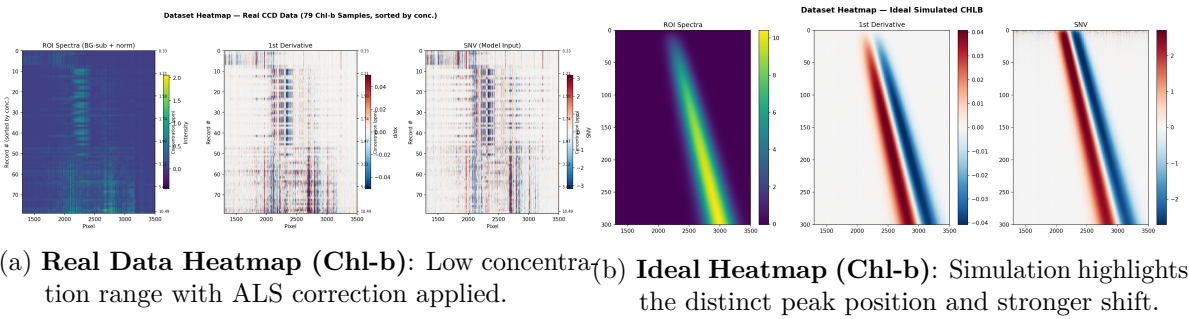
i Experimental Artifacts

The laboratory environment during initial calibration session was not fully rigid. Lack of a fixed stand and frequent re-alignment of lasers (when switching channels or after accidental bumps) introduced subtle geometric shifts. These fluctuations, along with slight repositioning of the cuvette, are clearly captured by the sensor’s sensitivity as observed in the heatmap’s jagged transitions. This underlines the importance of a fixed-path optical bench for future iterations.

Chl-a Heatmaps (Real vs. Ideal):



Chl-b Heatmaps (Real vs. Ideal):



7.4 Current Model & Data Assessment

Table 6 compares the current data collection conditions against the ideal parameters derived from the simulation study.

Table 6: Current data collection parameters versus the simulation-derived ideal case.

Parameter	Current Dataset	Simulation Ideal
Sample Count	~83 (Chl-a), ~79 (Chl-b)	200+ per analyte
Frame Averaging	10×	64–128×
Integration Time	Variable (240–5000 ms)	300–700 ms (standardised)
Outlier Removal	Manual	MAD + Hotelling T ²
PLS Components	9	4–5
CV Test R ²	0.85–0.87	0.997
RMSE	~1.5 ppm	0.29 ppm

Table 7: Summary of model investigation results compared against the simulation target.

Investigation	Physical Issue	CV R^2	RMSE (ppm)	Finding
Baseline PLS	—	0.867	~1.55	Funnel residuals (IFE)
Log-transform	Heteroscedasticity	~0.88	~1.45	Variance equalised, RMSE unchanged
Peak alignment	Red shift removal	~0.85	~1.60	Worse — peak position is a feature
Piecewise model	IFE non-linearity	~0.88	~1.40	Marginal gain, boundary instability

Investigation	Physical Issue	CV R^2	RMSE (ppm)	Finding
Simulation (PLS)	Ideal data (n=300)	0.997	0.29	5× better with proper data
Simulation (CNN)	Ideal + deep learning	0.999	0.20	Best achievable

7.5 Discussion of Data Quantity vs. Algorithm

Each investigation confirmed that the three physical phenomena affecting model performance — heteroscedastic Beer-Lambert response, concentration-dependent peak shift, and IFE saturation — are all *correctly handled* by the standard PLS preprocessing pipeline.

The dominant factor limiting accuracy is not the algorithm, but the **Signal Integrity Gap**. As seen in Figure 33, the real data suffer from a noise floor that obscures fine spectral features in the 1st derivative. Moving from 10x to 128x frame averaging (as recommended by simulation) will provide the SNR necessary for the 1D CNN to achieve its target < 0.3 ppm precision.

Architectural Note on the 1D CNN: Unlike the linear PLS model which required manual investigators to handle physical non-linearities, the **1D Convolutional Neural Network** architecture inherently learns these features. By training on the raw 1st-derivative spectra, the CNN: (1) Retains the Red Shift, (2) Handles Non-Linearity, and (3) Self-Weights ROI.

8 Conclusion & Future Outlook

8.1 Limitations

- **Concentration Drift:** Acetone evaporation and chlorophyll degradation during testing.
- **Instrumental Noise:** CCD electronic noise and sensitivity limits.
- **Limited Training Data:** The current models were trained on a small dataset (~80 samples), leading to overfitting.

8.2 Planned Improvements

- **Optical Upgrade — Diffraction Grating:** Upgrading from the current **600 lines/mm** grating to a **1200 lines/mm** grating would double spectral dispersion, enabling sub-nanometre resolution and improved separation of overlapping Chl-a and Chl-b fluorescence bands.

- **Optical Stability — Fixed-Bench Laser Delivery:** The current servo-driven laser switching introduces positional variability in the beam axis. Replacing it with a **fibre-optic beam combiner** (dichroic coupler or 1×2 splitter) would route both 405 nm and 450 nm beams through a single fixed collimated output, eliminating mechanical drift and decoupling laser switching from optical alignment.
- **Solvent Replacement — DMSO:** Acetone's high volatility (b.p. 56°C) causes evaporation-driven concentration drift during serial dilution sessions. **Dimethyl sulfoxide (DMSO, b.p. 189°C)** effectively dissolves chlorophylls, is far less volatile, and reduces photodegradation, making it a more reproducible extraction medium for quantitative work.
- **Phase 2 Combined Model:** As described in Section 5.4, the next algorithmic step is a dual-output model trained on both Chl-a and Chl-b simultaneously, using the existing UV-Vis calibrated dataset. The implementation is in `ml_model/train_combined_model.py`.
- **Simulation-Derived Data Collection Strategy:** The simulation study (Section 6) quantified exactly what data quality is needed to achieve $R^2 > 0.99$. The recommended protocol (Table 4) specifies: **200+ unique concentrations** with linear or $\sqrt{}$ spacing over 0.1–20 mg/L; **128× frame averaging** (currently 10×) to reduce noise by $\sim 3.5\times$ via $\sqrt{N}$ averaging; a **standardised integration time of 300–700 ms** (auto-exposure targeting 3000–3500 ADC counts) to eliminate exposure-driven variance; and **MAD + Hotelling T²** automated outlier rejection in place of manual review. Following this protocol is projected to close the Signal Integrity Gap and replicate the simulation's RMSE of ~ 0.29 mg/L.

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